Goldbach Midpoint Transitions and Twin–Cousin Rescue: A Computational Study to One Billion

Paul A. Bilokon

15 October 2026

Abstract

We study Goldbach representations about a moving pivot, distinguishing the loss of a direct continuation from the loss of all representations. For pivots congruent to 1 modulo 3, direct extinction has an exact primality criterion. Dirichlet's theorem then supplies infinitely many such pivots. We conjecture that each extinction is rescued by a twin pair and a prime pair at distance four, and show that this conjecture implies binary Goldbach and infinitely many twin primes. An equivalent four-prime convolution leads to a stronger residue-specific conjecture. An odd-only Eratosthenes sieve and a complete fallback search test every pivot through 10^9: among 284,220,751 residue-specific targets, no failure occurs, and no extinction pivot occurs in the other residue classes. A stricter right-hand orientation already fails at 277. We give exact algorithms, finite counts, code, and the logical limits of these observations; neither Goldbach nor the twin-prime conjecture is proved.

Keywords: Goldbach representations; twin primes; cousin primes; sieve; experimental number theory.

1 Fixed primes and moving pivots

Let $\mathbb{P}_{\mathrm{o}}$ be the set of odd primes. For $n \geq 3$, define the ordered representation set

$$\mathcal{G}_n = \{(p,q) \in \mathbb{P}_{\mathrm{o}}^2 : p+q = 2n\}, \qquad E_n = \{p \in \mathbb{P}_{\mathrm{o}} : 2n-p \in \mathbb{P}_{\mathrm{o}}\}. \tag{1}$$

Both (p,q) and (q,p) are included when $p \neq q$. A diagonal pair is included once. Let $g(n)$ denote the number of unordered representations, including a possible diagonal. We write G for the assertion $\mathcal{G}_n \neq \varnothing$ for every $n \geq 3$; together with $4 = 2+2$, this is binary Goldbach. We write T for the assertion that infinitely many odd twin-prime pairs exist.

Primality does not change when the pivot moves. Sieving identifies the fixed set of primes; the moving-pivot question concerns which two members of that set have the required midpoint. If $p = m-d$ and $q = m+d$, then relative to $n = m+h$ their signed positions are

$$p-n = -(d+h), \qquad q-n = d-h.$$

The second expression is a signed coordinate, not necessarily a nonnegative radius. If the fixed prime $q+2h$ exists, its signed position is $d+h$, so it is symmetric to p about n.

Proposition 1.1 (Exact transition identity)**.** *Fix $(p,q) \in \mathcal{G}_n$. For any integer a, put*

$$u = p+2a, \qquad v = q+2-2a. \tag{2}$$

Then $(u,v) \in \mathcal{G}_{n+1}$ if and only if $u, v \in \mathbb{P}_{\mathrm{o}}$. Every ordered representation at $n+1$ has this form relative to the fixed old pair.

Proof. The sum in (2) is $2n+2$. Conversely, given $(u,v) \in \mathcal{G}_{n+1}$, the integer $a = (u-p)/2$ gives the first coordinate, and subtraction of the two sums gives the second. □

The signed endpoint movements are $(2a, 2-2a)$. The values $a = 0, 1$ keep one endpoint fixed and increase the other by 2. The values $a = 2, -1$ give $(4, -2)$ and $(-2, 4)$. Larger movements are not excluded by the identity.

Proposition 1.2 (Transport from an earlier pivot)**.** *If $(p,q) \in \mathcal{G}_m$, $h \geq 1$, and $q+2h \in \mathbb{P}_o$, then $(p, q+2h) \in \mathcal{G}_{m+h}$. More generally, replacing the endpoints by primes $p-A, q+B$, for signed even integers A, B, advances the pivot by h exactly when $B - A = 2h$.*

Proof. The respective new sums are $2m+2h$ and $2m+B-A$. □

2 Direct extinction and the modulo-3 mechanism

Definition 2.1. An integer $n \geq 3$ is a *direct-extinction pivot*, denoted D_n, if

$$\mathcal{G}_n \neq \varnothing, \qquad \forall (p,q) \in \mathcal{G}_n : \quad p+2 \notin \mathbb{P}_o \text{ and } q+2 \notin \mathbb{P}_o. \tag{3}$$

This means that all direct continuations fail. It does not assert $\mathcal{G}_{n+1} = \varnothing$, and it does not mean that any old prime has become composite.

Let $\mathcal{T} = \{t \in \mathbb{P}_o : t+2 \in \mathbb{P}_o\}$, the set of lower twin members.

Proposition 2.2 (Exact twin characterization)**.** *For every $n \geq 3$,*

$$D_n \quad \Longleftrightarrow \quad E_n \neq \varnothing \text{ and } E_n \cap \mathcal{T} = \varnothing.$$

Proof. Every endpoint appears in E_n. An endpoint p has a direct continuation precisely when $p+2$ is prime, that is, when $p \in \mathcal{T}$. □

Proposition 2.3 (Modulo-3 extinction)**.** *Suppose $n \geq 4$, $n \equiv 1 \pmod 3$, and $\mathcal{G}_n \neq \varnothing$. Then*

$$D_n \quad \Longleftrightarrow \quad 2n-3 \text{ is composite.} \tag{4}$$

Proof. If $2n-3$ is composite, no representation involves 3. Every remaining prime endpoint is 1 or 2 modulo 3. As $2n \equiv 2 \pmod 3$, both endpoints must be $1 \pmod 3$. Adding 2 gives multiples of 3 greater than 3.

Conversely, if $2n-3$ is prime, the pair $(3, 2n-3)$ continues to $(5, 2n-3)$ at $n+1$. Since $2n-3 \geq 5$, it is either prime or composite, completing the equivalence. □

Proposition 2.4 (Allowed movements across the residue switch)**.** *Let $n \equiv 1 \pmod 3$. If $(p,q) \in \mathcal{G}_n$ and $(u,v) \in \mathcal{G}_{n+1}$ have all four endpoints greater than 3, their movements in (2) satisfy*

$$a \equiv 2 \pmod 3, \qquad (u-p, v-q) = (6k+4, -6k-2), \quad k \in \mathbb{Z}. \tag{5}$$

Proof. The old endpoints are $1 \pmod 3$, whereas $2n+2 \equiv 1 \pmod 3$ forces both new endpoints to be $2 \pmod 3$. Thus $2a \equiv 1 \pmod 3$, giving $a = 3k+2$. □

The cases $k = 0, -1$ are the twin-and-four-gap exchanges. The cases $k = 1, -2$ give $(10, -8)$ and $(-8, 10)$. Thus the residue switch alone does not require twins.

Theorem 2.5 (Infinitely many direct-extinction pivots)**.** *Every prime $n \equiv 19 \pmod{30}$ is a direct-extinction pivot. In particular, infinitely many direct-extinction pivots exist unconditionally.*

Proof. The pair (n, n) belongs to $\mathcal{G}_n$. Also $n \equiv 1 \pmod 3$, and $2n-3$ is a multiple of 5 greater than 5. Proposition 2.3 applies. Finally, $\gcd(19, 30) = 1$, so Dirichlet's theorem gives infinitely many prime values of n in this progression; see Sutherland [3, Theorem 17.1]. □

3 Twin-assisted replacement and its consequences

Conjecture 3.1 (Universal twin-assisted replacement, TR)**.** For every $n \geq 3$, if D_n, then some $(p, q) \in \mathcal{G}_n$ satisfies

$$p + 4 \in \mathbb{P}_{\mathrm{o}}, \qquad q - 2 \in \mathbb{P}_{\mathrm{o}}. \tag{6}$$

Ordering makes (6) include both geometric orientations. The primes $q-2, q$ are twins, and $p, p+4$ are separated by four. The new pair is $(p+4, q-2)$. Neither constituent prime pair is required to be consecutive in the full prime sequence.

Proposition 3.2 (Four-prime convolution)**.** *The replacement in Conjecture 3.1 at a given n exists if and only if*

$$C(n) := \sum_{\substack{t,c \geq 3 \\ t+c=2n-2}} \mathbf{1}_{\mathbb{P}_{\mathrm{o}}}(t)\mathbf{1}_{\mathbb{P}_{\mathrm{o}}}(t+2)\mathbf{1}_{\mathbb{P}_{\mathrm{o}}}(c)\mathbf{1}_{\mathbb{P}_{\mathrm{o}}}(c+4) > 0. \tag{7}$$

The sum is over odd integers. For each positive summand, $(t+2, c) \in \mathcal{G}_n$ and $(t, c+4) \in \mathcal{G}_{n+1}$.

Proof. Given (p, q) satisfying (6), set $t = q - 2$ and $c = p$. Then $t + c = 2n - 2$ and all four required entries are prime. Conversely, set $(p, q) = (c, t + 2)$ for any positive summand; then $(p + 4, q - 2) = (c + 4, t)$ is the new representation. □

Sorting the old pair puts the twin pair on either side. The cases $a = -1$ and $a = 2$ of (2) describe these two orientations. The representation (7) has no distinguished ancestor: a new Goldbach pair can be compared with multiple old pairs.

Conjecture 3.3 (Residue-specific four-prime rescue)**.** For every integer $n \geq 4$ with $n \equiv 1 \pmod 3$ and $2n - 3$ composite, $C(n) > 0$.

Unlike Conjecture 3.1, this asserts an old Goldbach pair rather than assuming one. Proposition 2.3 shows that it implies Conjecture 3.1 at all direct-extinction pivots in this residue class. It says nothing by itself about hypothetical extinction pivots in the other residue classes.

Proposition 3.4 (Local admissibility)**.** *For $n \equiv 1 \pmod 3$, the four forms in (7) have no local congruence obstruction. For $n \not\equiv 1 \pmod 3$, their reductions cover every residue modulo 3, except that an actual prime value equal to 3 can bypass this divisibility obstruction.*

Proof. Put $M = 2n - 2$ and regard t as the variable. For a prime r, the forms $t, t + 2, M - t, M - t + 4$ vanish at the residues $0, -2, M, M + 4 \pmod r$. At $r = 2$ one residue is forbidden. At $r = 3$, the forbidden set has two elements when $n \equiv 1 \pmod 3$ and three elements otherwise. For $r \geq 5$, at most four residues are forbidden, so at least one remains. □

In particular, surviving classes modulo every *fixed* product of primes do not guarantee a four-prime solution in the bounded interval $3 \leq t \leq 2n - 5$. A prime form equal to a small sieving prime must also be handled separately.

Proposition 3.5 (A twin-member representation at every step)**.** *If Conjecture 3.1 holds, every even integer at least 6 has a Goldbach representation with an endpoint belonging to a twin-prime pair.*

Proof. At $n = 3$ use $3 + 3 = 6$. Inductively, a direct continuation replaces a lower twin member t in an old pair by $t + 2$, so the new pair contains a twin member. At an extinction pivot, (6) supplies a new pair containing the lower twin member $q - 2$. □

This places the transition conjecture alongside the broader prime-plus-twin-member conjecture discussed by Pandey [1]. The latter does not require the old and new representations to share the four-prime configuration in (7).

Theorem 3.6. *The universal statement* TR *implies* G.

Proof. Induct from $\mathcal{G}_3 \ni (3,3)$. Suppose $\mathcal{G}_n \neq \varnothing$. If D_n fails, at least one endpoint of some old pair has a prime increment by 2, yielding a representation at $n+1$. If D_n holds, (6) yields one. Thus $\mathcal{G}_n \neq \varnothing$ for every $n \geq 3$. □

Lemma 3.7 (Finite endpoint exclusion)**.** *For any finite set $B \subset \mathbb{P}_o$, infinitely many prime pivots $n \equiv 19 \pmod{30}$ satisfy D_n and $B \cap E_n = \varnothing$.*

Proof. The empty-set case is Theorem 2.5. Otherwise write $B = \{b_1, \ldots, b_k\}$. Choose distinct primes $r_i > \max(5, b_i)$ and impose

$$n \equiv 19 \pmod{30}, \qquad 2n \equiv b_i \pmod{r_i} \quad (1 \leq i \leq k). \tag{8}$$

The Chinese remainder theorem gives a residue class A modulo $M = 30 \prod_i r_i$. It is coprime to M: its reduction modulo 30 is 19, and modulo each r_i it is the nonzero residue $2^{-1}b_i$. Dirichlet's theorem therefore supplies infinitely many prime $n \equiv A \pmod{M}$.

Restrict to those large enough that $2n - b_i > r_i$ for every i. Then $r_i \mid 2n - b_i$, so every $2n - b_i$ is composite. Hence $b_i \notin E_n$ for every i. Each such prime n satisfies D_n by Theorem 2.5. □

Theorem 3.8. *The universal statement* TR *implies* T.

Proof. Suppose there are only finitely many odd twin pairs and let B consist of all their upper members. Lemma 3.7 supplies a pivot with D_n but $B \cap E_n = \varnothing$. A replacement as in (6) requires q to be an upper twin member belonging to E_n, a contradiction. □

Remark 3.9. Infinitely many successful transitions alone would not force infinitely many distinct twin pairs: one fixed twin pair can participate at many pivots. Lemma 3.7 is the additional argument ruling out finite reuse under the *universal* replacement statement. The proof does not require twins on the right.

4 Directional versions and finite counterexamples

Definition 4.1 (Right-hand exchange, RTE_n)**.** The property RTE_n holds if some $(p, q) \in \mathcal{G}_n$ satisfies

$$p + 4, q - 2 \in \mathbb{P}_o, \qquad q - 2 > n + 1.$$

Both twin members $q - 2, q$ then lie strictly to the right of the *new* pivot.

The proposed universal implication $D_n \Rightarrow \mathsf{RTE}_n$ is false. Its first counterexample for $n \geq 3$ is $n = 277$.

Example 4.2 (The transition $277 \to 278$)**.** The old unordered pairs are

$$(7, 547), (13, 541), (31, 523), (67, 487), (97, 457), (157, 397),$$
$$(181, 373), (223, 331), (241, 313), (271, 283), (277, 277).$$

Here $277 \equiv 1 \pmod{3}$ and $551 = 19 \cdot 29$, so D_{277} holds. Among the old right endpoints, only $523, 313, 283$ have backward twins. Their corresponding replacements are

$$(35, 521), \qquad (245, 311), \qquad (275, 281),$$

all with composite left endpoints. Thus RTE_{277} fails. Nevertheless,

$$241 + 313 = 554 \quad \longrightarrow \quad 239 + 317 = 556$$

is a twin-assisted replacement using the left-hand twins $239, 241$.

A different directional property concerns the new representation, without requiring it to arise by the exchange in (6).

Definition 4.3 (Right-hand twin endpoint, RTP_n)**.** There exists $(u, v) \in \mathcal{G}_{n+1}$ with $v > n+1$ and either

$$v + 2 \in \mathbb{P}_{\mathrm{o}}, \quad \text{or} \quad v - 2 \in \mathbb{P}_{\mathrm{o}} \text{ and } v - 2 > n + 1.$$

Thus the right endpoint of a new pair belongs to a twin pair wholly right of the new pivot.

Conjecture 4.4 (Right-hand twin endpoint)**.** For every $n \geq 3$, $D_n \Rightarrow \mathsf{RTP}_n$.

Proposition 4.5. *Conjecture 4.4 implies both* G *and* T.

Proof. For Goldbach, use the induction in Theorem 3.6: a non-extinction step has a direct continuation, and at an extinction step RTP_n supplies a new pair. For twins, Theorem 2.5 gives unbounded extinction pivots, and RTP_n supplies twins with both endpoints greater than $n + 1$. Their endpoints are therefore unbounded. □

For example, at $n = 277$, the new pair $137 + 419 = 556$ has the right-hand twins $419, 421$. These give the next representation $137 + 421 = 558$. This establishes RTP_{277} but does not repair the failed implication to RTE_{277}.

5 What a converse from Goldbach would require

Conjecture 5.1 (Proposed conditional implication)**.** If binary Goldbach is true, then infinitely many twin primes exist: $\mathsf{G} \Rightarrow \mathsf{T}$.

This is recorded as a proposed implication, not as a consequence of the preceding theorems. The stronger proposed implication $\mathsf{G} \Rightarrow \mathsf{TR}$ would suffice. The proved implications are

$$\mathsf{TR} \Rightarrow \mathsf{G}, \qquad \mathsf{TR} \Rightarrow \mathsf{T}.$$

They do not reverse automatically. Goldbach only guarantees some integer a in Proposition 1.1; it does not restrict a to $-1, 0, 1, 2$.

Proposition 5.2 (Residual possibility under finitely many twins)**.** *Assume* G *and suppose there are only finitely many twin pairs. Then for every sufficiently large prime* $n \equiv 109 \pmod{210}$ *there exists an integer* $k \geq 1$ *such that*

$$n - (6k + 2) \in \mathbb{P}_{\mathrm{o}}, \qquad n + (6k + 4) \in \mathbb{P}_{\mathrm{o}}. \tag{9}$$

Proof. Such prime pivots are unbounded by Dirichlet's theorem. They satisfy $n \equiv 19 \pmod{30}$, so D_n holds. Also $n \equiv 4 \pmod{7}$, giving $7 \mid 2n - 1$; for these large pivots $2n - 1 > 7$. Thus no representation of $2n + 2$ involves 3.

By G, choose $u \leq v$ with $u + v = 2n + 2$. Both endpoints are $2 \pmod 3$. Since $n \equiv 1 \pmod 6$ and $u \leq n + 1$, parity and the residue condition give $u = n - (6k + 2)$ for an integer $k \geq 0$, and hence $v = n + (6k + 4)$. If $k = 0$, then $n - 2, n$ are twins. For n beyond all twin endpoints, this is impossible. Hence $k \geq 1$. □

Proposition 5.2 is a conditional consequence, not a contradiction. To prove Conjecture 5.1 by this route, one would need an additional result excluding (9) for at least one suitable pivot under the finite-twin assumption. No such result is proved here.

For illustration, $109 + 109 = 218$ can be replaced by $89 + 131 = 220$, with movements $(-20, 22)$. Neither 89 nor 131 has a twin neighbour, since $87, 91, 129, 133$ are composite. Other representations at this pivot do involve twins. The example shows that a particular successful replacement need not use twins; it does not refute any proposed universal implication between the classical conjectures.

6 Earlier pivots, six-step transport, and pooled counts

Lemma 6.1 (Exclusion of some crossing-out primes)**.** *Suppose $(p,q) \in \mathcal{G}_{n-h}$ and a prime r divides $q+2h$. If $r \mid h$, then $r = q$. In particular, if $q > \sqrt{2n}$ and $q+2h$ is composite, its least prime factor is at most $\sqrt{2n}$ and does not divide h.*

Proof. The two divisibilities imply $r \mid q$, so primality gives $r = q$. A composite $q + 2h = 2n - p < 2n$ has a prime factor at most $\sqrt{2n}$, excluding that exception when $q > \sqrt{2n}$. □

Conjecture 6.2 (Six-step transport)**.** For every $n \geq 7$, there exists $p \in \mathbb{P}_{\mathrm{o}}$ such that

$$p + 6 \in \mathbb{P}_{\mathrm{o}}, \qquad 2n - 6 - p \in \mathbb{P}_{\mathrm{o}}.$$

This transports a representation at $n-3$ to one at n. It implies G after the finite cases below 14 are checked. The lower limit is intentional: at $n = 6$, the sole old pair is $3 + 3 = 6$ and both increments by 6 equal 9.

Proposition 6.3 (Local admissibility of six-step transport)**.** *Set $M = 2n - 6$. For a prime r, the residues of p making at least one of $p, p+6, M-p$ divisible by r are*

$$0, -6, M \pmod{r}.$$

Their number ν_r is

condition	ν_r
$r = 2$	1
$r = 3,\ 3 \mid n$	1
$r = 3,\ 3 \nmid n$	2
$r \geq 5,\ r \mid n(n-3)$	2
$r \geq 5,\ r \nmid n(n-3)$	3

In particular, $\nu_r < r$ for every prime r.

Proof. For $r = 2, 3$, the residues $0, -6$ coincide. For $r \geq 5$ they differ, and the third residue coincides with one of them exactly when $M \equiv 0$ or $M \equiv -6$, equivalent to $r \mid n-3$ or $r \mid n$. This gives the table. □

CRT consequently gives $\prod_{r \leq z}(r - \nu_r) > 0$ surviving classes in a full period modulo $\prod_{r \leq z} r$. It does not guarantee a survivor in $3 \leq p \leq M - 3$. Moreover, prime values equal to a sieving prime must be treated as exceptions when translating divisibility avoidance to primality.

For $H \geq 1$, define

$$w_H(r) = \#\{h : 1 \leq h \leq H,\ r - 2h \in \mathbb{P}_{\mathrm{o}}\}, \tag{10}$$

$$T_H(n) = \sum_{\substack{3 \leq r \leq 2n-3 \\ r \in \mathbb{P}_{\mathrm{o}}}} w_H(r) \mathbf{1}_{\mathbb{P}_{\mathrm{o}}}(2n - r). \tag{11}$$

A positive summand supplies a transport from $n - h$ to n.

Proposition 6.4 (Full-window identity)**.** *For $n \geq 4$,*

$$T_{n-3}(n) = \sum_{(p,r) \in \mathcal{G}_n} (\pi(r) - 2), \tag{12}$$

and $T_{n-3}(n) > 0$ if and only if $\mathcal{G}_n \neq \varnothing$.

Proof. For $3 \le r \le 2n-3$, every odd prime below r is of the form $r-2h$ with $1 \le h \le n-3$. There are $\pi(r)-2$ such primes, proving (12). Positivity implies a representation. Conversely, any representation of $2n \ge 8$ has an endpoint $r \ge 5$, whose predecessor 3 gives a positive weight. □

For instance, $g(100) = 8$ but $T_{97}(100) = 370$. Different earlier states can feed the same target pair. This identity explains why abundance of available gaps alone is not an independent lower bound for representations at a specified pivot.

7 Sieve accounting and related results

For $n \ge 3$, let $L_n = \mathbb{P}_o \cap [3, n]$. For each odd prime $r \le \sqrt{2n-3}$, define

$$A_r(n) = \{p \in L_n : 2n - p \text{ is composite with least prime factor } r\}.$$

Proposition 7.1 (Exact disjoint sieve accounting)**.**

$$g(n) = \pi(n) - 1 - \sum_{\substack{r \le \sqrt{2n-3} \\ r \in \mathbb{P}_o}} |A_r(n)|. \tag{13}$$

Every prime divisor of an odd composite $q \le 2n-3$ is less than n.

Proof. Each $p \in L_n$ contributes exactly one unordered representation if its complement is prime; otherwise it belongs to exactly one set $A_r(n)$. There are $\pi(n)-1$ possible left primes. Every composite complement has a least prime factor at most $\sqrt{2n-3}$, proving (13). Finally, if $r \mid q$ and q is odd composite, q/r is an odd integer at least 3, so $r \le q/3 < n$. □

Thus the fact that all crossing-out primes lie left of the pivot is included in the exact accounting. Positivity still requires a strict bound on their collective coverage. No such bound is deduced here from earlier representations.

Two related results motivate the formulations. First, if $g_1 < g_2 < \cdots$ are the even integers representable as sums of two primes, the result attributed to Baker, Harman, Jia and Pintz, as presented in Pintz [2, Section 14], gives

$$g_{j+1} - g_j \ll g_j^{21/800}.$$

It controls the distance to another successful pivot, but does not supply a representation at the next pivot or an aligned twin pair.

Second, the paired Jacobsthal function of Ziller and Morack [4] measures the longest obstruction to simultaneous coprimality along paired progressions. Their Proposition 3.2 proves a sufficient implication to Goldbach from a conjectured bound $h_2(k) < p_k^2 - p_k$, where $h_2(k) = j_2(p_k\#)$. That bound is not established in their paper. Reflecting $(n-d, n+d)$ to $(d-n, d+n)$ preserves divisibility and converts the symmetric sieve into a paired progression. This is a structural connection, not a proof of the replacement conjectures above.

8 Reproducible finite computations

The independent dependency-free script `verify.mjs` constructs a dense Eratosthenes sieve through 2,000,010. It checks every integer pivot through 10,000 using explicit old and new representation sets, retaining diagonal pairs and both exchange orientations. It separately checks the residue-specific conjecture through 10^6 by scanning *cousin* bases c and testing $t = 2n-2-c$. Its machine-readable output is `verification.json`.

The standalone C++17 program `goldbach_rescue_scan.cpp` uses one bit per odd integer in an Eratosthenes sieve through $2N + 2$, where N is the user-selected pivot bound (default 10^9). It checks *every* $4 \le n \le N$ as follows:

(i) If $n \equiv 1 \pmod 3$ and $2n - 3$ is composite, search for $C(n) > 0$, without first assuming $\mathcal{G}_n \neq \varnothing$.

(ii) If $n \not\equiv 1 \pmod 3$, search for a direct continuation by enumerating lower twin members t and testing $2n - t$ for primality. If none exists, check explicitly whether $\mathcal{G}_n \neq \varnothing$. Only then is n a nonmodular extinction target requiring a quartet.

(iii) For a quartet, try precomputed twin bases $t \le 10^6$ and cousin bases $c \le 10^6$. If neither works, scan the *entire* remaining interval $t, c > 10^6$, testing the four prime values. Thus a finite cutoff cannot generate a false counterexample.

The boundary cases 3 and 5 are handled by direct primality tests, not discarded by the modulo-3 argument. The program stops at the first genuine failure and prints its pivot, a check for an old representation, and a representation of the next even number if one exists. With four threads the bit sieve occupied 119.2 MiB at $N = 10^9$; the particular run recorded in `verification_billion.txt` took 31.28 seconds including sieve construction. Timings depend on hardware and load.

Check	Result
Direct-extinction pivots, $3 \le n \le 10{,}000$	2,197
Failures of TR among them	None
Failures of RTE_n among them	$277, 676, 1012, 1342$
Failures of RTP_n among them	None
Six-step transport, $7 \le n \le 1{,}000{,}000$	No failures in 999,994 cases

The 10,000-pivot figures above come from the direct representation-set enumerator. The following larger-range figures are cumulative output of the separate C++ program:

Maximum pivot N	Residue-specific targets	Other extinctions	Rescue failures
10^6	258,814	0	0
10^7	2,697,854	0	0
10^8	27,793,376	0	0
10^9	284,220,751	0	0

At 10^9, the first-stage small-pattern search found a quartet for every target, so there were no exhaustive fallbacks. The largest number of small candidates examined at one pivot was 2034. The one-million-pivot result is independently reproduced by `verify.mjs`, which uses a different, byte-per-integer sieve and cousin-first enumeration. Neither computation proves the universal conjectures.

To repeat the billion-pivot experiment on a machine with a C++17 compiler and roughly 120 MiB available for the sieve, use:

```
g++ -O3 -march=native -std=c++17 -pthread \
  goldbach_rescue_scan.cpp -o goldbach_rescue_scan
./goldbach_rescue_scan 1000000000 --threads 4 \
  --progress 250000000 --samples 8
```

The range is inclusive and the output prints both the number of candidate pivots and any counterexample. The source accepts `-mode residue` or `-mode general` to test either formulation alone, as well as `-small-bound` to exercise the exhaustive fallback. The absence

of extinction at $n = 3, 4$ means that counting from 5 instead gives the same total. The largest least working p in the six-step search was 2281, at $n = 940132$:

$$2281 + 1877977 = 1880258, \qquad 2287 + 1877977 = 1880264.$$

The first six observed extinction transitions admit the following right-hand exchanges:

Pivots	Old pair	New pair
$19 \to 20$	$7 + 31$	$11 + 29$
$34 \to 35$	$7 + 61$	$11 + 59$
$40 \to 41$	$7 + 73$	$11 + 71$
$49 \to 50$	$37 + 61$	$41 + 59$
$61 \to 62$	$13 + 109$	$17 + 107$
$64 \to 65$	$19 + 109$	$23 + 107$

Example 4.2 explains why this directional pattern does not persist universally.

9 Logical status and remaining questions

The exact identities, modulo-3 propositions, infinitude of direct extinction, finite endpoint exclusion, four-prime equivalence, local admissibility, and conditional implications from the replacement conjectures have proofs in this note. The universal claims TR, Conjecture 3.3, RTP, and six-step transport remain conjectural. The universal right-hand exchange claim is disproved by finite examples.

The proposed implication $\mathsf{G} \Rightarrow \mathsf{T}$ remains unproved here. Under a hypothetical finite set of twins, the CRT argument eliminates twin-assisted exchanges at infinitely many extinction pivots; it does not eliminate all larger-gap exchanges. Establishing that at least one of those pivots admits no larger-gap replacement would complete a contradiction, but that assertion is an additional missing result, not a consequence of the modular analysis.

The principal proved conditional conclusion is

$$\boxed{\mathsf{TR} \implies \mathsf{G} \text{ and } \mathsf{T}.}$$

It formalizes the proposed rescue mechanism while keeping its unproved existence requirement explicit.

References

[1] Tushar Pandey. An intermediate conjecture between Goldbach and Dubner: Every even number is the sum of a prime and a twin prime. arXiv:2608.02381 [math.NT], 2026. URL https://arxiv.org/abs/2608.02381. Preprint; the prime-plus-twin-member statement remains conjectural.

[2] János Pintz. Landau's problems on primes. *Journal de Théorie des Nombres de Bordeaux*, 21(2):357–404, 2009. doi: 10.5802/jtnb.676. URL https://www.numdam.org/item/JTNB_2009__21_2_357_0/.

[3] Andrew V. Sutherland. Dirichlet characters and primes in arithmetic progressions. MIT 18.785 Number Theory I, Lecture 17, 2015. URL https://math.mit.edu/classes/18.785/2015fa/LectureNotes17.pdf. Theorem 17.1; lecture dated 10 November 2015.

[4] Mario Ziller and John F. Morack. Divisibility in paired progressions, Goldbach's conjecture, and the infinitude of prime pairs. arXiv:1706.00317 [math.NT], 2017. URL https://arxiv.org/abs/1706.00317.